Density bins and overflow

Density bins count cells on a grid and sum overflow above capacity

The idea

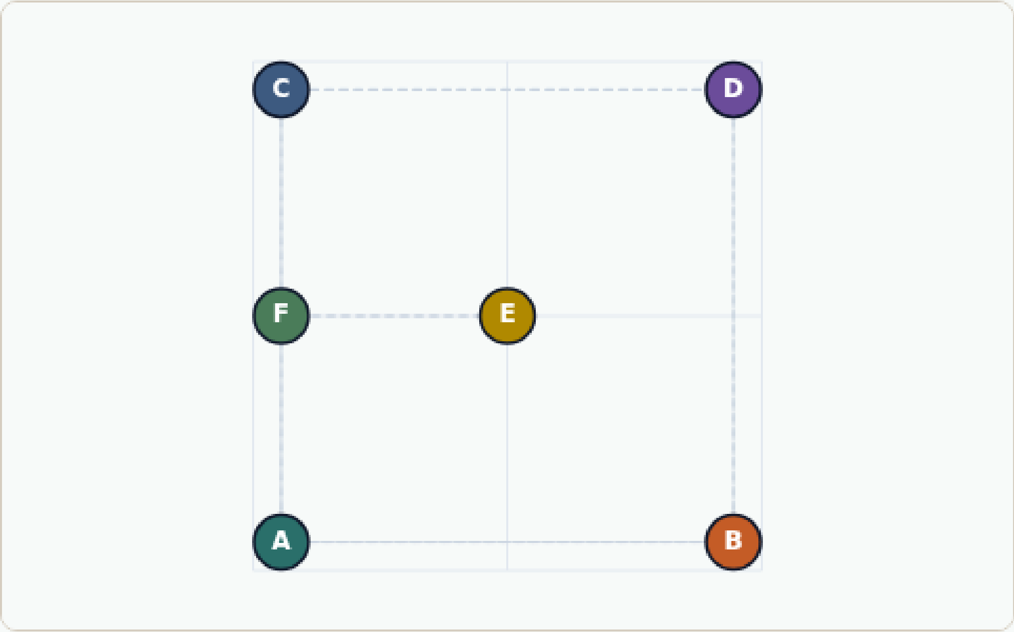
- Assign each cell to a bin by coordinates
- Report HPWL and overflow together
- A pretty wirelength with piled bins fails the density half of placement
- <!-- algorithm-walkthrough -->

Partition the die into bins

DENSITY BINS

Partition the die into bins

Step 1 / 5 · grid-idea · module03-01-density-bins



Explanation

Density bins count cells on a regular grid. Default teaching grid is two-by-two over $[0,8]\times[0,8]$. Capacity one means each bin may hold one cell before overflow.

- Assign cell → bin by (x,y)
- overflow = Σ max(0, count − cap)
- Report with HPWL

```
Grid: 2×2
Capacity: 1
World: [0,8]²
```

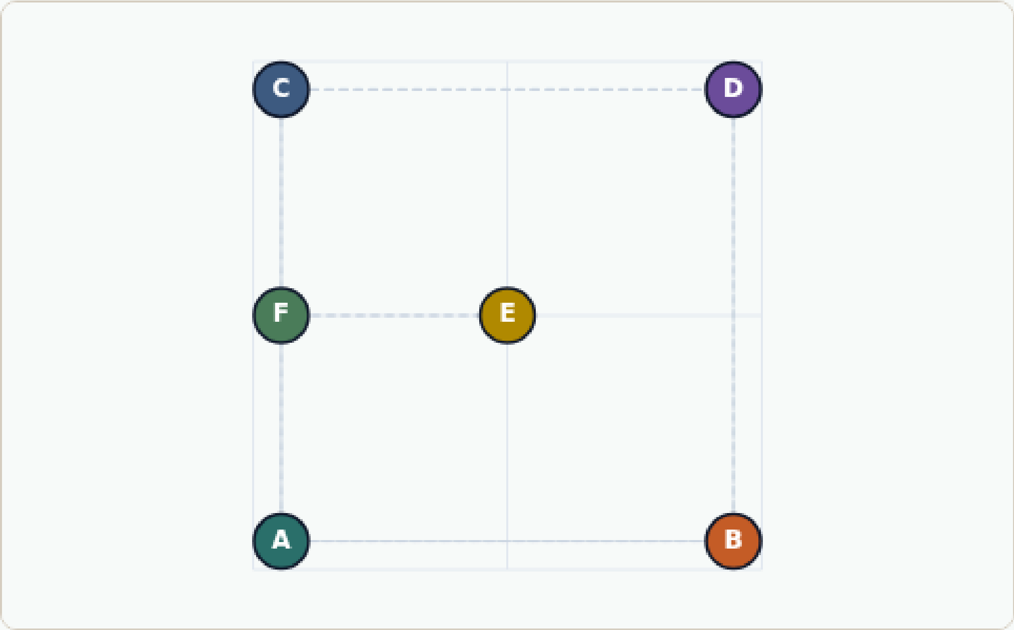
Partition the die into bins

Starter overflows by two

DENSITY BINS

Starter overflows by two

Step 2 / 5 · starter-overflow · module03-01-density-bins



Explanation

On the spread starter with capacity one, two bins hold two cells each. Overflow sums to two even though HPWL is already fifty-two.

- Bottom-left and bottom-right busy
- overflow = 2 at cap 1
- Spread ≠ legal density

```
overflow: 2
counts: [[1,1],[2,2]]
HPWL: 52
```

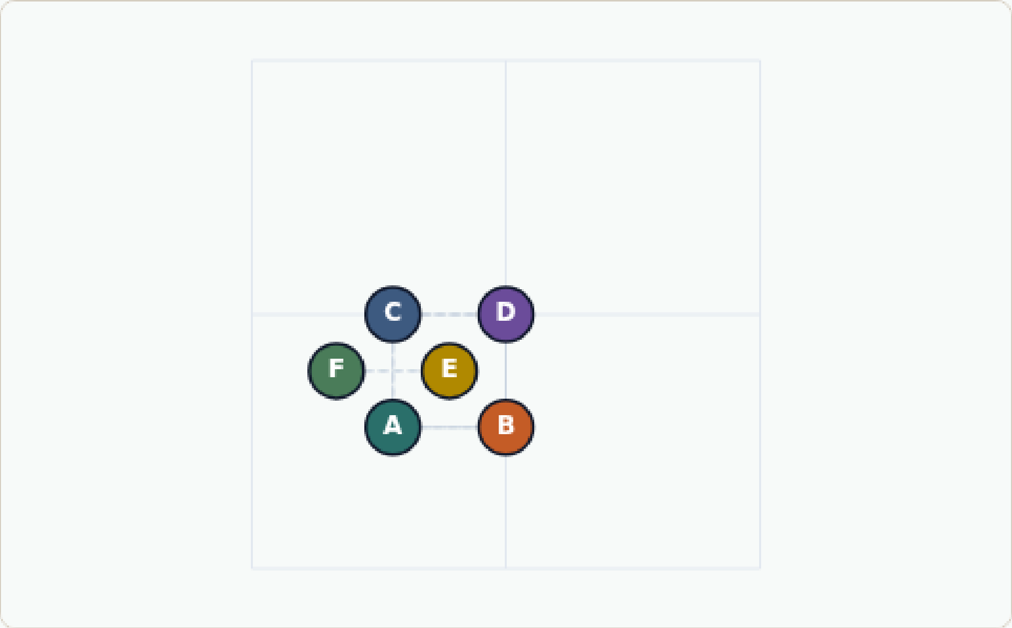
Starter overflows by two

Golden also overflows at cap one

DENSITY BINS

Golden also overflows at cap one

Step 3 / 5 · golden-still-overflow · module03-01-density-bins



Explanation

Compact HPWL fourteen still piles three cells into one bin. Overflow remains two at capacity one—pretty wirelength is not automatic legality.

- Golden HPWL = 14
- Same overflow 2 at cap 1
- Density is a separate objective

```
overflow: 2
counts: [[3,1],[1,1]]
HPWL: 14
```

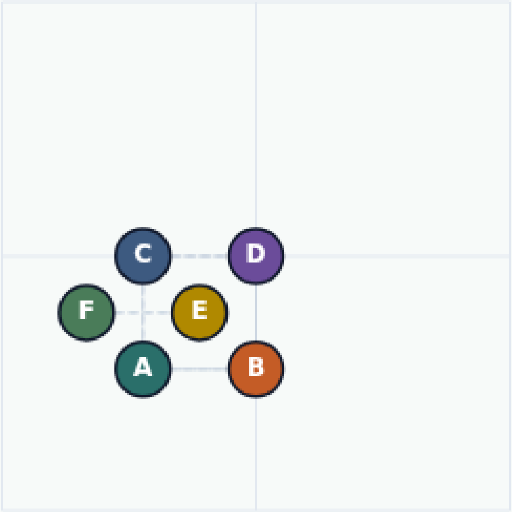
Golden also overflows at cap one

Raise capacity to ease overflow

DENSITY BINS

Raise capacity to ease overflow

Step 4 / 5 · raise-capacity · module03-01-density-bins



Explanation

With capacity two on the golden placement, overflow drops to one. Capacity is part of the spec—quote it with the overflow number.

- cap 1 → overflow 2
- cap 2 → overflow 1
- Same coordinates, new budget

```
density2x2Cap2GoldenOverflow: 1
Measured: 1
```

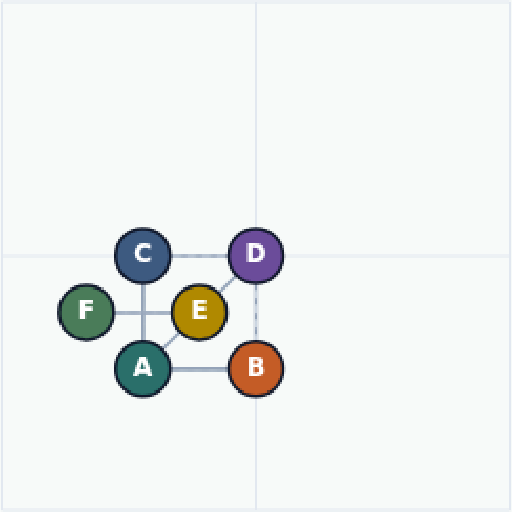
Raise capacity to ease overflow

HPWL and density travel together

DENSITY BINS

HPWL and density travel together

Step 5 / 5 · takeaway · module03-01-density-bins



Explanation

Always report wirelength and bin overflow. A placement that wins HPWL while stacking bins fails the density half of the story.

- Starter & golden: overflow 2 @ cap 1
- cap 2 golden: overflow 1
- Next: spread / legalize lite

```
starter overflow@1: 2
golden overflow@1: 2
golden overflow@2: 1
```

HPWL and density travel together

Browser lab track

- In the browser lab track, open the **density-bins** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module’s examples and the course `common/` solvers
- Parse `tiny_place.json`, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you’re ready, take the short quiz, then continue to the next module